Determining hysteresis, remanence and coercive fields of a ferromagnet

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In this paper different ferromagnetic core materials were used to plot their hysteresis curves and determine their magnetic behavior. With those curves it was possible to identify characteristic properties like coercivity, saturation magnetisation, core loss and remanence. For both iron-nickel alloyings PERMENORM 5000Z and PERMENORM 5000H2 the following values are measured the magnetisation saturation is $B_{sat,5000H2} = (1.4441 \pm 0.0079)\,\mathrm{T}$ and $B_{sat,5000Z} = (1.428 \pm 0.066)\,\mathrm{T}$, the remanence $B_{rem,5000H2} = (0.6603 \pm 0.0034)\,\mathrm{T}$ and $B_{rem,5000Z} = (1.428 \pm 0.066)\,\mathrm{T}$ and the coercivity $H_{coer,5000H2} = (3.63 \pm 0.29)\,\frac{A}{m}$ and $H_{coer,5000Z} = (14.6 \pm 1.1)\,\frac{A}{m}$. The core loss was not measured for the PERMENORM 5000Z in contrast to the core loss $\xi_{cf} = (2.62 \pm 0,45)\frac{m*J}{kg}$ for the mentioned PERMENORM 5000H2. Additionally the remanence, coercitivity and core loss result in $B_{rem,ST37K} = (0.547 \pm 0.012)\,\mathrm{T}$, $H_{coer,ST37K} = (358 \pm 14)\,\frac{A}{m}$ and $\xi_{cf,ST37K} = (0.264 \pm 0.045)\ \frac{J}{kg}$. It was not possible to calculate the saturation magnetisation because of the limited effective range. Lastly ferromagnets are compared with wood a non magnetic material. For this material a linear relation of B and H was recorded. For wood the permeability was calculated to $\mu_r = 1.835 \pm 0.024$.

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1 Introduction

Ferromagnets are very important nowadays as they act as if they had a memory contrary to para- and diamagnets. Thus they are used for electromotors of cars or other machines or harddiscs. They are utilized as datastore because of their characteristics as remanence and coercitivity which are explained in 3. Their hysteresis curve does not follow a function of B(H) like dia- and paramagnets as it depends on the state it was in before (magnetised or not) and the magnetic field, which effects the ferromagnet. Quantities like remanence, coercive fields and saturation magnetisation are specific for the material. Hence we determine quantities like those by measuring hysteresis curves in the following paper and compare them to wood as a non-magnetic material.

2 Theory

First of all it is important to clear some terms and definitions in relation to the theory of these following experiments. As mentioned before ferromagnets are used to save information in long-term like electromotors of cars. But there are also other types of magnets, which can´t be used for saving data for a longer period of time. To compare some attributes of ferromagnets and diamagnets, wood is used as an example of diamagnetism. Diamagnets, other than ferromagnets are not able to construct a magnetic field without an outer magnetic induction. They create an inner magnetic field antiparallel to the outter one. As wood is only slightly diamagnetic, this effect cannot be observed properly. So if there is an exterior magnetic field which tries to put information inside the diamagnet, all of these pieces of information are gone when the outer field is turned off because of the material structure.

Ferromagnets are able to build their own magnetic field without exterior magnetic induction. That's because of the material structure of ferromagnets. The magnetic moments of the electrons tend to align parallel. So if there is an outer magentic field the moments inside the magnet orient themselves towards the direction of the exterior one. This happens in a macroscopic scale. In a minor dimension,the microscopic one, so called magnetic domains direct themselves. Magnetic domains are regions in a magnetic material where the spins of electrons couple with each other and point to the same direction. While the outside magnetic field gets stronger, these domains get

bigger and their borders migrate. So more and more magnetic moments allign themselves with the outer field. Those borders are called domain walls. If there is no outside influence there are different material specific magnetic domains in one substance and the domain walls are steady which influence characteristic sizes like remanence, coercitivity and core loss. So if the outer magnetic field H increases and therefore the domain walls expand in the ferromagnet, the magnetic field B increases. This relation is shown in so called hysteresis curves. Regarding to this curves there are a few terms, which should be explained beforehand. If the voltage range is big enough, the saturation magnetisation can be observed. If the saturation magnetisation occours all magnetic domains are alligned parallel to the outer field. When hitting this spot the magnetic field B is only increasing with the incline of the vacuum permeability μ_0. If now the outer magnetic field is reduced constantly some of the magnetic domains start to change their orientation because of thermal fluctations and the magnetic flield decreases. This happens on a different trajectory in H-B-space than before because the thermal fluctation is weaker than the spin interaction in ferromagnetic materials at room-temperature. So if the point $H = 0$ is passed there is some magnetisation left, which is called remanence. After that, the direction of the outer magnetic field is inverted and the field is increased again. Eventually the curve gets to the spot where the magnetic flux is completely gone. The value of H at this point is called coercitivity. Now the procedure is the same as before but with a reversed magnetic field. The hysteresis curve is complete if the saturation magnetization from the beginning is achieved again. So every hysteresis curve is point-symmetrical.

When the magnetic domains shift in a different direction, energy is lost in form of heat. This heat is proportional to the area inside the hysteresis curve and can be calculated with the formula:

$$\xi = \frac{E}{m} = \frac{1}{\rho} \oint B dH \tag{1}$$

3 Experiment

In the following experiments, hysteresis curves of different materials were taken with the experiment setup given in fig. [1]. The installation consists of two toroidal inductors, both of them are wind around the material to be analysed. The primary coil generates the outer magnetic field which induces a magnetisation in the examined material and a voltage and current in the induction coil. The generated magnetic field strength H is given by the voltage drop U_R measured with a known resistance R with the formula

$$H = \frac{N_1 U_R}{\pi \frac{1}{2} (d_i + d_a) R} \tag{2}$$

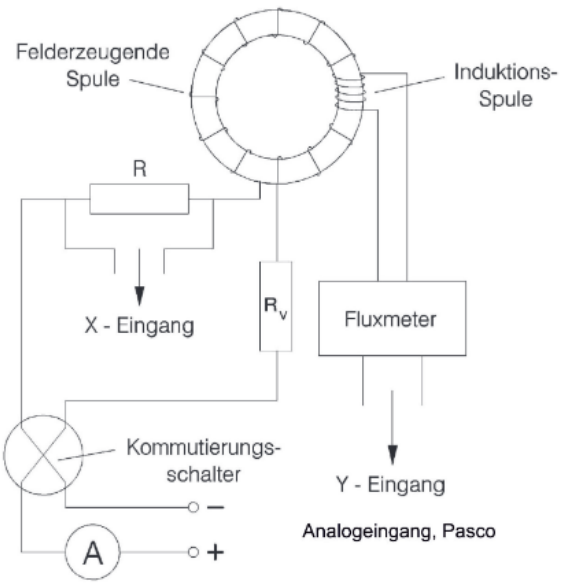

Fig. 1: Build up of the following experiment to plot hysteresis curves of different materials. The basic construction stays the same only the two given resistances R_v and R and the material of the core change in some sections.

with N_1 the winding number, d_i and d_a the inner and outer diameter of the field creating inductor. A resistance with the value $R = (66,7 \pm 1,4)\,\Omega$ is used for measuring at small currents. Whereas for higher currents the measuring resistor $R_m = 0.20\,\Omega$ is utilised. With the change of the flux density a voltage surge in the induction coil is created. Via the built in fluxmeter, this voltage surge can be integrated against time to get a voltage which is proportional to the flux density B calculated by the formula:

$$B = \frac{\gamma_f U_f}{N_2 A_{eff}} \tag{3}$$

Where γ_f is an proportionality factor of the fluxmeter U_f and the fluxmeter output voltage. N_2 is the winding number of the induction coil. A_{eff} the area which is actual saturated by the magnetic field and given by

$$A_{eff} = \frac{1}{2}(d_a - d_i)\, h\, f \tag{4}$$

with the diameters d_a and d_i, the height h and the bulk factor f of the core. With this setup it is possible to plot the hysteresis curves using a PASCO Universal Interface 550 Analog-Digital-Wandler“. However, because of the functionality of the fluxmeter based on using integration with operational amplifiers, all values are shifted in the direction of the flux density.

To get symmetric plots, those data points have to be centered with:

$$U_c = U_B - \frac{1}{2}(U_{min} + U_{max}) \tag{5}$$

Whereas U_c is the centered voltage of the induction coil and U_B the induced voltage in it. U_{min} and U_{max} are the lowest and highest measured voltages. Because there is no infinite input resistence plugged in the fluxmeter, there is also a linear shift in time, regarding to the data points of B. How this disruptive factor is adjusted is explained in 5.1 on the basis of the hysteresis curve of wood.

3.1 Measurement with PERMENORM 5000H2

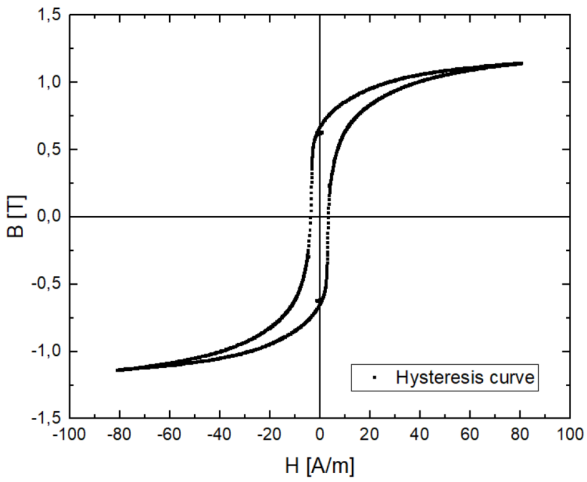

Fig. 2: Hysteresis curve of the alloy PERMENORM 5000H2

In this first experiment the hysteresis curve of the core PERMENORM 5000H2 is plotted. The material is an iron-nickel alloy. The resistance $R = 66.80 \pm 0.34\,\Omega$ is used to regulate the electrical current below $I_{max} = 100\,\text{mA}$. The procedure to receive the hysteresis curve is described in sec. 3. The recieved and corrected hysteresis curve is shown in figure [2]. By means of the hysteresis curve, characteristic quantities like remanence and coercitivity can be determined. To obtain those, a linear fit is placed at the intercepts as shown in fig. [4]. The data is presented in chart [1]. Since the voltage range creating the magnetic field H is too narrow to measure the saturation voltage, the hysteresis curve is recorded again with a current range of $I_{max} = 2.3\text{V}$ using a resistance of $R = 0.2\,\Omega$. The resistance is treated without an uncertainty as its error is negligible compared to the other ones. The resulting curve is shown in fig. [3]. The saturation magnetisation is ascertained by the help of a linear

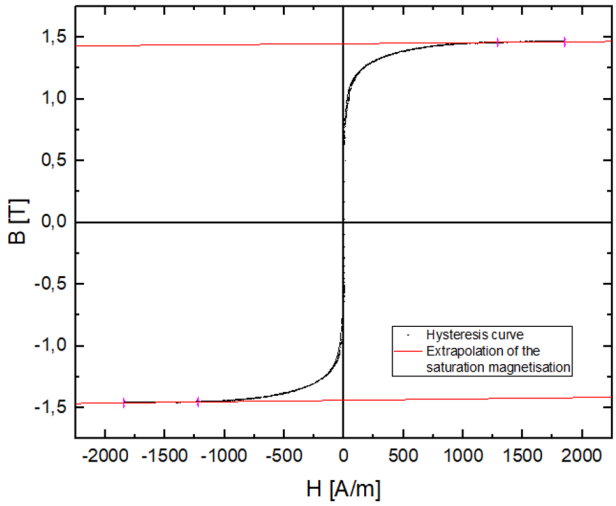

Fig. 3: Hysteresis curve of PERMENORM 5000H2 with extended range to determine the saturation magnetisation, which is calculated with the help of extrapolation.

fit and its intercept with the B-axis. The resulting quantities are portrayed in tab. [1]. Additional the core loss was calculated with numerical integration of the given hysteresis curve and also inserted in tab. [1].

References regarding saturation magnetization and coercivity state the value of these characteristics for PERMENORM 5000H2: $B_{saturation,Lit} = 1.55$ T and $H_{coercivity,Lit} = 5\ \frac{A}{m}$. In this experiement the values of the searched characteristics do not agree with literature values. This could be due to the fact that the saturation magnetisation was not reached or that the process of manufacturing has an influence on the magnetisation characteristics.
This material can be used for a transformer due to the small coercivity since it is very easy and energy-efficient to change the poles. Due to this fact this material is unsuitable for long lasting permanent magnets and their use or storing data.

$B_{rem}\,[mT]$	$H_{coer}\,[\frac{A}{m}]$	$B_{sat}\,[mT]$	$\xi_{cf}\,[\frac{mJ}{kg}]$
660.3±3.4	3.63±0.29	1,444.1±7.9	2.62±0.45

Tab. 1: Calculated values for the alloying PERMENORM 5000H2

3.2 Flat steel type ST37K

In figure [4] the received hysteresis curve of flat steel is shown. The core of the coil is of the type ST37K. Again, by the intercept points of curve and axes we get the values of remanence B_{rem} and coercivity

$H_{coercivity}$. Since it was not possible to get the saturation magnetisation based on the too small voltage range creating the magnetic field H in the inducter coil, there is no value measured in this section. The core loss ξ_{loss} in this experiment is calculated through numerical integration of the area in the curve with formula (1).

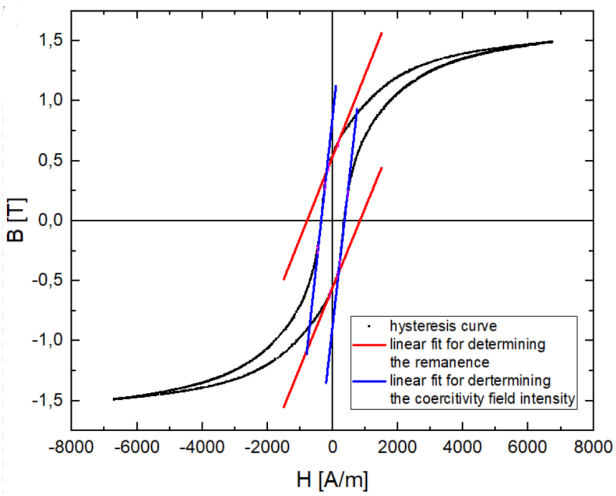

Fig. 4: Hysteresis curve of a steel core of the type ST37K. Remanence and coercivity were found by the help of the shown reference lines.

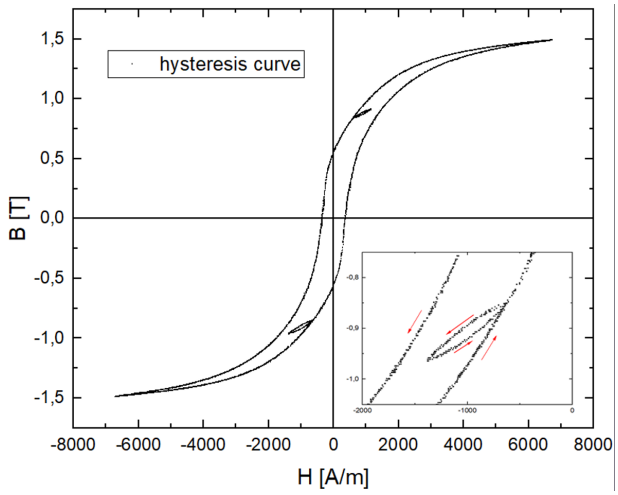

Fig. 5: Hysteresis curve of a steel core type ST37K with additional loops. One of these loops is shown in a close-up picture in the bottom-right-hand corner. The direction of the hysteresis curve is indicated with arrows.

The change of direction of the graph is analysed in addition to the normal hysteresis curve in figure [5]. Therefore the normal procedure plotting the curve was done, but after reaching the starting point a second round was started. There are two loops shown

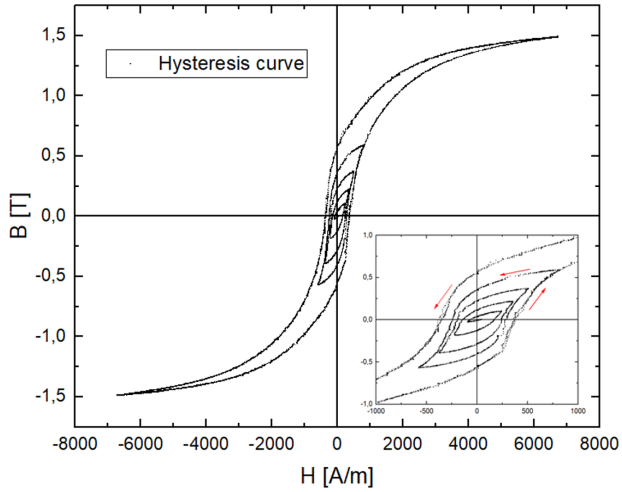

Fig. 6: Plotted hysteresis curve of steel core type ST37K. This curve shows the demagnetisation of the material in form of decreasing loops converging to zero. It is visible that the point of zero magnetisation is not reached exactly in the close-up picture of the inner area on the bottom-right-hand corner.

in fig. [5]. They result by decreasing/increasing H contrary to the direction of the curve before for a short period of time. B follows now a different track than before because of the hysteresis effect explained in **??**. After that the magnetic field is now increased/decreased according to the original direction.
In fig. 6 the same effect was used to demagnetise the core: Before reaching the saturation, the magnetig field strength H is decreased again and the process is repeated such that the magnetic flux tends to zero. After this the outer magnetic field is increased until the starting point is hit again, so the resulting plot is a closed hysteresis curve. The actual execution was a little bit hard as the off-set had to be considered as well as the shift, Hence a small remanence is remaining. But compared with the earlier determined remanence in fig. [4] this one is way smaller which is indicating that this method works very well. Due to the high coercive field, the steel core ST37K is not so well used for transistors but storing data.

B_{rem} $[T]$	H_{coer} $[\frac{A}{m}]$	B_{sat} $[T]$	ξ_{cf} $[\frac{J}{kg}]$
0.547 ±0.012	358 ±14	-	0.264±0.045

Tab. 2: Calculated values for the material flat steel of the type ST37K

3.3 Measurement with the core PERMENORM 5000Z

PERMENORM 5000Z is another iron- nickel alloying. The characteristic quantities are obtained identically to the other core with an alloy. The values of remanence and saturation magnetisation are given in table [3]. Because there is no data given for the density of this material it is not possible to calculate ξ_{loss}.
Like PERMENORM 5000H2 this core with the alloy 5000Z can also be classified as a magnetically soft material, which is used for short time memory in transistors because of the small coercivity and core loss as described before.

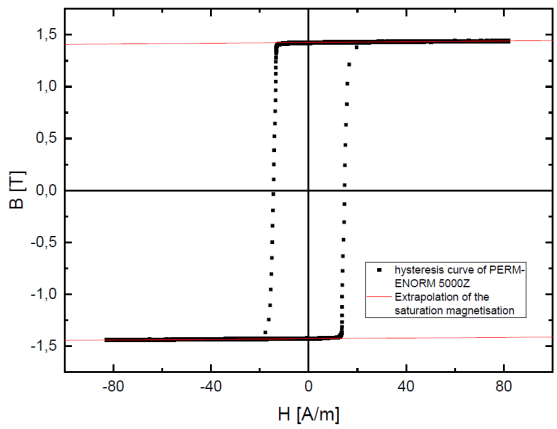

Fig. 7: Hysteresis curve of PERMENORM 5000Z. Out of this plot all values of the materia can be calculated.

B_{rem} $[T]$	H_{coer} $[\frac{A}{m}]$	B_{sat} $[T]$	ξ_{cf} $[\frac{J}{kg}]$
1.428 ±0.066	14.6 ±1.1	1.428 ±0.066	-

Tab. 3: Calculated values for the alloying PERMENORM 5000Z

3.4 Wooden core for comparison

The course of action is again the same. The B(H)-curve of wood is plotted and shown in figure [8]. This material was selected to be able to compare a ferromagnetic substance with a non ferromagnetic, slightly diamagnetic one. In the image the linear correlation of B and H is nicely visible. Also it is recognisable that the inner magnetic field is diminutive because wood is not magnetic so there is no induced field possible. This is due to the inner structure of wood. The atom spins do not connect with each other as strong as in ferromagnets so it is impossible to build up magnetic domains. Because these bonds are so weak the thermal fluctation is relatively strong and destroys any parallelism after decreasing the outer field. Therefore, there is no hysteresis curve nor a left remanence at $H = 0\,\frac{A}{m}$. The slope of the curve is exactly the permeability $\mu = \mu_0 * \mu_r$. We receive the permeability of wood $\mu_r = 1.835 \pm 0.024$ with $\mu_0 = 1.25663706212(19) * 10^{-6}\ \frac{N}{A^2}$. According to this result wood is paramagnetic, whereas according to literature wood has a permeability μ_r<1. Wood is a slightly diamagnetic material. The reason for this discrepancy is expected in the experiment set-up, as all the coils are stacked in a shelf right next to each other, so that the magnetic fields of the cores interfer. We assume that the ferromagnetic cores above and below sophisticate the permeability μ_r of wood. Moreover the shelf is made out of steel which contributes to the distortion.

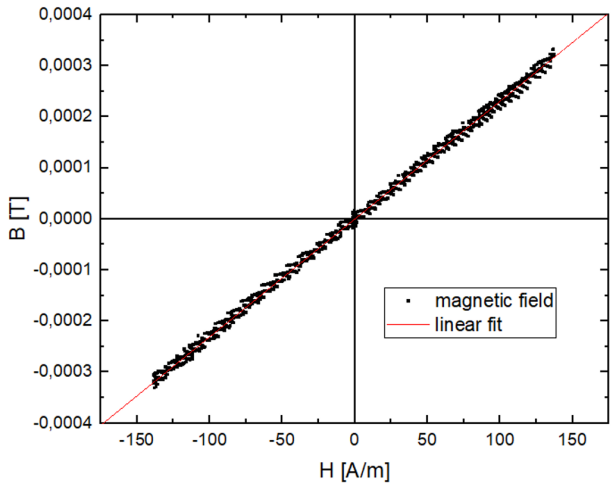

Fig. 8: Plot of a wooden core which shows a linear behavior and no hysteresis curve.

4 Results

In this paper three different ferromagnets and one theoretical diamagnet were analysed. With the help of plotted hysteresis curves it was possible to calculate characteristics like remanence, coercivity, saturation magnetisation and specific energy loss, which are all specific for those materials. To be able to compare the behaviour of ferromagnets with non magnetical substances, wood was examined as well. All values of the individual cores are listed beneath: For the ST37K steel core we got for remanence $B_{rem} = (0.547 \pm 0.0012)$T, for coercivity $(385 \pm 14\ \frac{A}{m})$ and the core loss with the value $\xi_{cf} = (0.264 \pm 0.045)\ \frac{J}{kg}$. For the two iron- nickel cores we got for PERMENORM 5000Z the remanence $B_{rem} = (1.428 \pm 0.066)\ T$, the coercivity $H_{coer} = (14.6 \pm 1.1)\ \frac{A}{m}$ and the saturation magnetisation with the value $B_{sat}(1.428 \pm 0.066)\ T$. For the

other one with the allying PERMENORM 5000H2 we got $B_{rem} = (660.3 \pm 3.4)$T, $H_{coer} = (3.63 \pm 0.29)\ \frac{A}{m}$, $B_{sat}(1,444.1 \pm 7.9)\ T$ and at least $\xi_{ef} = (2.62 \pm 0.45)\ \frac{mJ}{kg}$. For the wooden core which shows a linear behavior it was possible to measure the permeability $\mu_r = (1.835 \pm 0.024)$.
All curves are corresponding to the theoretical expectations. Additionally the turn-around in the hysteresis curve was detected with the steel ST37K core. With this technique it was possible to demagnetise the core and to plot the received curve.

Literatur

[1] Meschede, Gerthsen Physik, 23, 2005

[2] Frauenfelder und Huber, Einführung in die Physik II,

[3] Physikalisches Grundpraktikum, Universität Würzburg, Modul C2, Versuch 39

[4] https://vacuumschmelze.com/products/soft-magnetic-materials-and-stamped-parts/40-to-50-Nickel-Iron

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5 Appendix

5.1 Drift of the fluxmeter

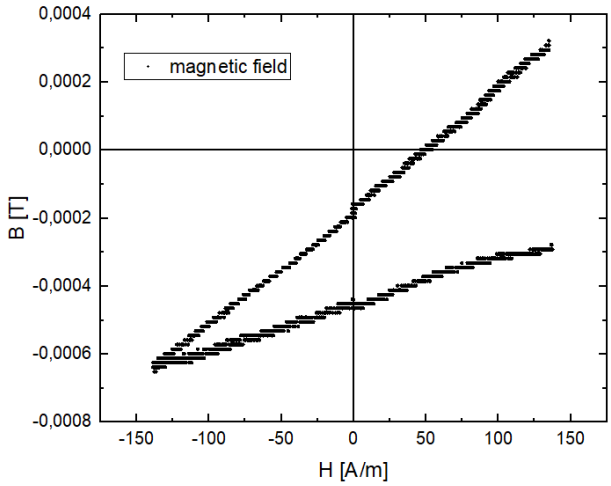

Fig. 9: Magnetic characteristic curve without adjusting the fluxmeter drift

In fig. 9 the plot of the wooden core with the fluxmeterdrift is presented where only the B- offset is aligned. The flux density is drifting towards lower values with a seemingly constant rate wherefore the shift is assumed to be linear. Hence the final plot is not closed and does not hit the origin as would be expected. With the following formula it is possible to calculate the drift of the fluxmeter for each datapoint and hereby adjust the plot correctly

$$\dot{B}_{drift} = (\frac{\partial B}{\partial t})_{drift} \approx \frac{B_n - B_1}{t_{total}} \tag{6}$$

by the use of the first recorded flux density B_1 and the last one B_n and the duration of the whole testseries t_{total}. So this received time shift was added to all datapoints to get the adjusted final plot.

5.2 Comparing different methods of analysing core loss

The core loss was calculated numerically by integrating $\xi_{cf} = \frac{1}{\rho} \oint BdH$ in 3.1 and 3.2 . In the past, when computers were not invented, other methods had to be used to evaluate the data. For this reason we want to compare numerically evaluation by a computer and the manually one. We printed the hysteresis curve of the core 5000H2 and weighed a piece of paper and the cut out hysteresis curve with METTLER AJ150. First we calculated the area which is encircled by the hyyteresis curve:

$$(\int U_B\, dU_A)_{hyst} = (\int U_B\, dU_A)_{sq} \frac{A_{sheet}}{A_{sq}} \frac{m_{hyst}}{m_{sheet}} \tag{7}$$

The coordinate system in which the hysteresis curve was visible was not covering the whole DINA4 sheet, so we had to calculate the relation of the surface area of the whole DINA4 sheet A_{sheet} and the area of the coordinate system multiplied with the relation of the weighed mass of the hysteresis curve m_{hyst} and of the whole DINA4 sheet m_{sheet} and the integral of the encircled square of the coordinate system $(\int U_B \, dU_A)_{sq}$. U_B and U_A are the measured voltages of the magnetic field B and magnetic field strength H.
Since we need to calculate the core loss we need to calculate ξ_{cf} according to formula [1], we multiplied a factor α containing all the needed coil data of the core 5000H2 in 5.4 plus the one of the inducter coil. With the given quantities in tab. [4] we receive $\xi_{cf} = 3.3 \pm 1.3 \frac{mJ}{kg}$. The numerically calculated one amounts to $\xi_{cf} = 2.62 \pm 0.45 \frac{mJ}{kg}$. Those two optimum values of the core losses differ from 27.4%. Although the abberation of the optimum values seems pretty huge, those two values coincide including their uncertainties. The huge uncertainty of the manually evaluated method results from a pretty big uncertainty in reading the scala on the printed sheet and that we had to cut out the hysteresis curve in order to weigh it.
As we see, manually evaluating works but is less precice than evaluating numerically.

5.3 Errors

The given values at the workplace are assumed to be accurate and without error. So it is possible to calculate σ_B and σ_H with gaussian error propagation and the formulas [2] and [3].
The error of the fluxmeter is estimatet because the manufacturer information is missing. We set 0.5% of range as the value of the error. For the other used devices Voltcraft 333 and the interface, we took the errors $\sigma_{U,interface} = 1.22\,\mathrm{mV}$ and $\sigma_{VC333} = \pm(0.5\% + 2\mathrm{digits})$ from the moments.

5.3.1 Uncertainty of remanence and coercitivity

As there is an uncertainty in H and B there is like a uncertainty square around every data point. This is why we calculated the uncertainty for the remanence and coercitivity by multiplying one error with the associated tangent slope comparing those in e.g. fig [4]:

$$\sigma_{rem} = \sigma_B + \sigma_H \, m_{rem} \tag{8}$$

and:

$$\sigma_{coer} = \sigma_H + \frac{\sigma_B}{m_{coer}} \tag{9}$$

Here the tangent slope m_{rem} portrays the slope average of the tangents at the remanence. Analogous it is apllied for m_{coerc}.

5.3.2 Uncertainty of the saturation magnetisation

As the saturation magnetisation is determined by applying a tangent to the saturation, the uncertainty is calculated by linear regression.

5.3.3 Uncertainty of core loss

Since the resetting is calculated by numerically integrating the area of the hysteresis curve, we used the average of the relative uncertainty of H and B and accounted them by gauss.
For manually evaluating, we used reading uncertainties und accounted them by gauss as well.

5.4 coil data

	Fe-Ni H2	Fe-Ni Z	flat steel	wood
d_a	260mm	260mm	260mm	260mm
d_i	220mm	220mm	220mm	220mm
N1	605	610	1010	1010
N2	80	56	53	250
h	30,0mm	30,0mm	30,0mm	30,0mm
f	90%	90%	100%	100%
R	0,20Ω	0,20Ω	0,20Ω	0,20Ω

Tab. 4: values of the different coils used in this experiment